# Yuhang *Zhang*

PhD candidate · Synthetic organic & organometallic chemistry · Asymmetric catalysis · Medicinal chemistry

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## PROFILE

PhD candidate in **synthetic organic and organometallic chemistry**. Hands-on expertise across **complex molecule synthesis**, **asymmetric catalysis and chemical method development**, and **polymer chemistry**, supported by air-/moisture-free Schlenk technique and full-spectrum analytical characterization. Chemical Engineering BS provides foundations in process and plant design. Targeting **medicinal chemistry, synthesis chemist / R&D, and process development** roles in U.S. pharma and biotech.

## EDUCATION

### University of Houston

Ph.D., Chemistry

08/2022 – Expected 2027 · GPA 3.722 / 4.000

**Passed Oral Research Proposal**

### Technion — Israel Institute of Technology

(China campus, Shantou)

B.S., Chemical Engineering

09/2018 – 08/2022 · GPA 85.4 / 100 (Rank 19 / 74)

Dean's List (Top 15%) · Third Class Chancellor's Scholarship

## CORE SKILLS

### SYNTHESIS

Multi-step total synthesis · asymmetric catalyst synthesis & optimization ($Rh_2$(II) carboxylate) · organometallic complex assembly · organocatalysis (R-TN1) · medicinal chemistry (warfarin derivatives, flavone scaffold synthesis) · reaction condition screening & gram-scale scale-up · polymer synthesis & characterization.

### CHEMICAL ENGINEERING FOUNDATIONS (BS BACKGROUND)

UniSim plant design · integral process design (methanol-plant capstone) · process flow diagrams · mass / energy balance · technical reporting.

### AIR- & MOISTURE-FREE

Schlenk line · freeze-pump-thaw · cannula transfer · solvent / amine

## RESEARCH EXPERIENCE

### May Group · Department of Chemistry, University of Houston — 01/2023 – Present

*Doctoral Researcher · Advisor: Prof. Jeremy A. May · Houston, TX*

- **Dirhodium(II) catalyst — lead project.** Lead contributor on a 2nd-generation chiral bidentate $Rh_2$(II) biscarboxylate catalyst (initial design by Dr. H. Barzegar) featuring α-quaternary stereocenters and a fused-cyclohexane architecture; m-xylene tether bridges both Rh centers at ~90° bite angle for enantioselective carbene cascade reactions. Synthesized the full ligand and currently assembling the metal complex.
- **7-step total synthesis** of the m-xylene-tethered bidentate ligand; key intermediates carried through at hundreds-of-milligrams to gram scale.
- **Reaction optimization & scale-up:** scaled $PdCl_2$/PMHS ketone reduction from 100 mg to 750 mg at **~100% isolated yield** by tuning reagent dispersion and a 23–40 °C water-bath thermal profile; screened >10 conditions (atmosphere, MeOH grade, PMHS equivalents, temperature).
- **Air- / moisture-free workflows:** Schlenk line, freeze-pump-thaw degassing (reproduced at gram scale across multiple runs), DIPA distillation over $CaH_2$, low-temperature (−78 °C) Li-amide chemistry, and inert-atmosphere reagent transfer.
- **Medicinal chemistry side-projects:** Synthesized warfarin derivatives for ORP via R-TN1-organocatalyzed conjugate addition, demonstrating enantioselective organocatalysis on a medicinally relevant scaffold; gram-scale flavone scaffold synthesis (~1.4 g × 4 batches), with an arylboronic-acid-driven analog series planned for biological-activity studies.
- **Characterization:** multinuclear NMR ($^1$H, $^{13}$C, COSY, HSQC, HMBC); extensive chiral HPLC method screening for ee determination; GC-MS, MALDI-TOF, IR, X-ray; supported computational analysis of the Rh-carbene pocket.

### Texas Center for Superconductivity · University of Houston — 01/2024 – 09/2024

*Doctoral Researcher (joint) · Advisors: Prof. Jeremy A. May & Prof. Zhifeng Ren*

- Synthesized and characterized novel **fluorinated poly(arylene piperidinium) hydroxide** candidates aimed at anion exchange membrane (AEM) applications for water electrolysis.
- Investigated structure–viscosity relationships across polymer formulations; characterized polymer structure and end-group identity by **NMR and MALDI-TOF**.

### Chemical Engineering · Technion (China campus) — 10/2021 – 02/2022

*Project Lead, Chemical Integral Process Design · Shantou, China*

- Led a multi-member team to optimize a large-scale **methanol production process**; built the full PFD and mass / energy balance in **UniSim** to deliver a profitable plant design; authored the technical report and presented to faculty reviewers.

### Chemical Engineering · Technion (China campus) — 08/2021 – 01/2022

*Polymer Lab Student Assistant*

- Extruded polyethylene formulations with varied stabilizer loadings; quantified thermal stability with **DSC and TGA**; ran tensile testing across multiple polymer specimens and fabricated specialty-shaped specimens for downstream testing.

distillation (DIPA / $CaH_2$, THF) · low-temp Li-amide chemistry (−78 °C).

### ANALYTICAL

NMR (1D & 2D) · Chiral HPLC · GC-MS · MALDI-TOF · X-ray · IR · TGA · DSC

### COMPUTATIONAL & DATA

ChemDraw · Python · Pandas · Jupyter · MATLAB.

## CERTIFICATIONS

**Hewlett Packard Enterprise Data Science Institute, University of Houston:**
Selected Topics in AI · Intro to Deep Learning · Intro to Machine Learning · Principles of Data Management

## HONORS

Dean's List, Technion (Top 15%)
Third Class Chancellor's Scholarship, Technion (Top 15%)

### Chemical Engineering · Technion (China campus) — 08/2019 – 06/2021

*Undergraduate Research Assistant*

- Synthesized the ligand of a **photo-controlled organometallic catalyst** for olefin block-copolymer polymerization; tracked amidation and hydroboration reactions by MALDI-TOF, TLC, and NMR; co-translated a research proposal for an NSFC funding application.

## TEACHING & LEADERSHIP

### Teaching Assistant, Organic Chemistry Lab I · University of Houston

09/2022 – Present

- Instruct undergraduate laboratory sections in foundational organic techniques (distillation, recrystallization, extraction, TLC, NMR interpretation); grade lab reports; hold weekly office hours; scheduled for **6+ consecutive semesters**.

## PUBLICATIONS

### Acknowledged Contributions · Published Reviews — 2020

- Light-mediated olefin coordination polymerization and photoswitches.
- Catalytic dearomative hydroboration of N-heteroarenes.
- Transition-metal-free hydroelementation (E = B, Si, Ge, Sn) of alkynes.

## PRESENTATIONS & POSTERS

### Poster · TexSyn Symposium, University of Houston — 2026

- “Design and Synthesis of a Novel Chiral Bidentate Dirhodium(II) Catalyst for Enantioselective Carbene Cascade Reactions.” *Y. Zhang, H. Barzegar, J. A. May.*

### Poster · TexSyn Symposium, University of Houston — 2024

- “Warfarin Derivatives Efficiently Synthesized Enantioselectively via Organocatalyzed Conjugate Addition and Synthesis of Robust Anion Exchange Membranes (AEMs).” *Y. Zhang, C. Yuan, Z. Ren, J. A. May.*

### Graduate Seminar · Department of Chemistry, UH — 2024

- “Ni-Catalyzed Electrochemical Decarboxylative Cross-Coupling Reactions.”